E Additional Baselines

We compare RetrievalAttention with additional baselines InfiniGen and Quest on the RULER benchmark and show the results on Table 9. InfiniGen and Quest exhibit a noticeable drop in model accuracy compared to full attention. In contrast, RetrievalAttention performs best and achieves nearly the same accuracy as full attention across two benchmarks.

Table 9: Performance (%) of different methods in 128K context length on RULER.

	Methods	Act. Tokens	S1	S2	S3	M1	M2	M3	MQ	MV	VT	CW	FW	Q1	Q2	Avg.
Llama-3	FullAttention	128K	100.0	100.0	100.0	98.0	98.0	73.0	94.5	97.0	87.0	1.0	72.2	58.5	44.5	78.7
	InfiniGen	2K	99.0	91.5	24.5	82.5	25.0	0.0	30.3	27.8	67.3	1.2	45.5	33.0	32.5	43.1
	Quest	2K	100.0	100.0	98.5	98.5	36.5	0.0	48.9	64.3	89.4	1.0	64.5	45.0	39.5	60.5
	Ours	640 + 100	100.0	100.0	100.0	99.0	98.0	45.0	92.8	93.0	88.0	1.1	49.3	60.5	44.5	74.7

F Dynamic Retrieval Budget Allocation

We investigated the impact of adjusting the retrieval budget according to the sparsity degree across layers, by adopting the budget allocation policy from PyramidIKV (Cai et al., 2024). Specifically, we compare the performance of the original RetrievalAttention with and without the PyramidKV-based budget allocation strategy on the InfiniteBench benchmark, as shown in RTable 2. Specifically, for the original RetrievalAttention, we set a fixed budget of 2000 tokens for all heads in all layers. In contrast, PyramidKV dynamically adjusts the retrieval size across different layers, allocating more in lower layers and less in higher ones.

The results in Table 10 shows that PyramidKV allocation strategy achieves better performance in Retr.KV tasks, though it slightly decreases performance in the En.QA task. On average, the accuracy slightly surpasses that of the original RetrievalAttention. This indicates that dynamic budget allocation is promising but may require task-specific allocation strategies.

Table 10: Performance (%) of RetrievalAttention and RetrievalAttention w/ PyramidKV in 128K context length.

Methods	Retr.N	Retr.P	Retr.KV	Code.D	Math.F	En.QA	En.MC	Avg.
Full Attention	100.0	100.0	17.5	19.0	39.5	9.1	68.0	50.4
RetrievalAttention	100.0	100.0	14.5	18.5	40.0	8.7	67.5	49.9
RetrievalAttention w/ PyramidKV	100.0	100.0	16.0	18.5	40.0	8.5	67.5	50.1

G Performance on the larger model

To demonstrate the generalizability of our methods on larger models, we evaluated our method on Llama-3-70B-262k using a server with eight 40GB A100 GPUs by partitioning the model by layers across GPUs. We choose the most complex task KV retrieval in ∞-Bench to stress test the efficiency of RetrievalAttention and other baselines.

The results in the Table 11 shows that RetrievalAttention achieves nearly the same task accuracy as the exact KNN method Flat, and outperforms Quest by 80%. The decoding speed of RetrievalAttention is $3.5\times$ faster than Flat as it effectively reduces the vectors to scan.

Table 11: Performance (%) and decoding latency (s) in Llama-3-70B Model.

	Full	StreamingLLM	Quest	Flat	RetrievalAttention
Accuracy	35.0	0.0	13.0	24.0	23.5
Decoding latency	248	0.14	1.36	5.68	1.62